

Distributed HPC Workload & Power Grid Simulation Project

1. Project Overview

This project focuses on analyzing the impact of Machine Learning (ML) and High-Performance Computing (HPC) workloads on energy consumption and electrical grid behavior. The current system simulates how GPU-intensive workloads influence data center power demand and evaluates their effect on electrical infrastructure using power flow simulations.

The primary goal of the enhanced project is to transform the existing simulator from a static, assumption-based framework into a more realistic, data-driven, and optimization-oriented system.

2. Problem Statement

Modern ML workloads such as training and inference require significant computational power and heavily depend on GPUs. As AI adoption grows, data centers consume increasing amounts of electricity, creating challenges related to:

- High operational costs
- Peak power demand
- Grid stability
- Carbon emissions
- Energy-efficient workload scheduling

Current simulation approaches often rely on simplified assumptions and average power models, which fail to capture realistic time-varying workload behavior and infrastructure-level impacts.

This project aims to address these limitations through improved workload modeling, grid simulation, and optimization strategies.

3. Existing System (Already Implemented)

The current simulator already includes several foundational components.

3.1 GPU Workload Trace Processing

- GPU power traces are collected from ML workloads
- Time-series workload behavior is analyzed
- Training and inference traces are processed

3.2 Load Modeling

The simulator converts GPU-level traces into:

- Node-level power profiles
- Facility-level load estimates
- Aggregated HPC center demand

Currently, several assumptions are used, such as:

- Constant PUE (Power Usage Effectiveness)
- Simplified node power estimation
- Synthetic infrastructure behavior

3.3 Grid Simulation

The project uses:

- Synthetic grid models
- SimBench benchmark grids
- AC power flow simulations using pandapower

The simulator evaluates:

- Voltage levels
- Transformer loading
- Line loading
- Power flow convergence

3.4 Operational Scenarios

The current implementation supports:

- Baseline workload execution
- GPU power limiting
- PUE efficiency scenarios
- Workload shifting
- Load distribution scenarios

3.5 Dashboard

A Streamlit-based dashboard is available for:

- Running simulations
- Selecting parameters
- Visualizing outputs

4. Current Limitations

Although the existing system provides a strong foundation, several limitations still exist.

4.1 Synthetic Assumptions

A significant portion of the infrastructure behavior is assumption-based, including:

- Constant PUE
- Simplified node power
- Static facility behavior
- Simplified reactive power estimation

4.2 Limited Real-World Integration

The current system does not yet integrate:

- Real electricity prices
- Carbon intensity data
- Dynamic environmental conditions
- Renewable energy profiles

4.3 No Intelligent Decision-Making

The simulator currently evaluates scenarios but does not:

- Optimize workload scheduling
- Dynamically balance loads
- Make energy-aware decisions

5. Proposed Enhancements

The enhanced version of the project will focus on realism, optimization, and distributed systems behavior.

5.1 Improved Data Modeling

The system will incorporate additional real-world or estimated data such as:

- CPU utilization and power
- Memory utilization
- Storage and I/O activity
- Network traffic
- Dynamic node-level power

- Dynamic PUE behavior

5.2 Facility-Level Enhancements

The simulator will improve facility modeling by:

- Adding dynamic cooling behavior
- Improving rack-level aggregation
- Modeling load-dependent infrastructure efficiency

5.3 Distributed Multi-Center Simulation

The project will support multiple simulated HPC/data centers with:

- Distributed workload execution
- Load balancing strategies
- Multi-center coordination
- Regional energy analysis

5.4 Optimization and Scheduling

The enhanced simulator will include:

- Workload scheduling strategies
- Peak-load reduction mechanisms
- Cost-aware execution
- Load balancing optimization

5.5 Sustainability Analysis

The project will include:

- Electricity price integration
- Carbon intensity analysis
- Energy vs cost trade-off evaluation
- Sustainability metrics

5.6 Dashboard Improvements

The Streamlit dashboard will be extended with:

- Real-time visualizations
- Multi-scenario comparison
- Optimization result displays
- Interactive analysis tools

6. Data Required for Realistic Simulation

To reduce assumptions and improve realism, the following data will be integrated wherever possible.

Compute / Workload Data

- GPU power consumption
- GPU utilization
- CPU utilization and CPU power
- RAM / memory usage
- Storage and I/O activity
- Network traffic
- Workload metadata

Infrastructure Data

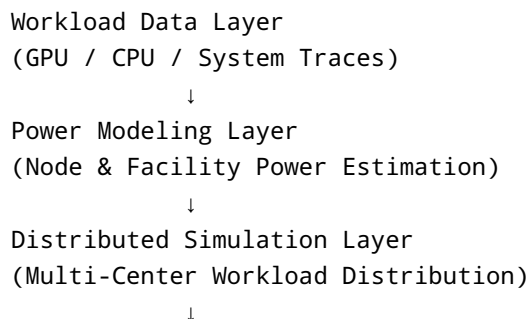
- Total node/server power
- Rack-level power usage
- Cooling system power
- Dynamic PUE values
- Environmental temperature

Grid & External Data

- Electricity prices
 - Carbon intensity data
 - Grid demand profiles
 - Renewable energy availability
 - Voltage and transformer loading data
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7. System Architecture

High-Level Architecture



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Grid Simulation Layer
(AC Power Flow using pandapower)
    ↓
Optimization Layer
(Scheduling & Load Balancing)
    ↓
Evaluation & Analytics Layer
(Cost, Energy, Carbon, Grid Metrics)
    ↓
Visualization Dashboard
(Streamlit Interface)
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8. Workflow

Step 1 — Workload Collection

Collect and preprocess workload traces:

- GPU usage
- Power traces
- Utilization patterns

Step 2 — Power Modeling

Convert workload activity into:

- Node-level power
- Facility-level power
- Aggregated HPC demand

Step 3 — Distributed Simulation

Distribute workloads across:

- Multiple nodes
- Multiple simulated centers
- Different operating scenarios

Step 4 — Grid Simulation

Run AC power flow simulations:

- Voltage analysis

- Transformer loading
- Line utilization
- Peak demand analysis

Step 5 — Optimization

Apply scheduling and balancing strategies:

- Load balancing
- Peak reduction
- Cost optimization
- Sustainability-aware scheduling

Step 6 — Evaluation

Evaluate:

- Energy consumption
- Operational cost
- Carbon emissions
- Grid stability
- Scalability

Step 7 — Visualization

Present outputs through:

- Interactive dashboards
- Scenario comparisons
- Real-time visualizations

9. Project Timeline

Weeks 1–2

- Data preprocessing & analysis
- Power and node-level modeling

Weeks 3–4

- Facility-level modeling
- Grid simulation improvements

Weeks 5–6

- Multi-center simulation
- Load balancing & scheduling optimization

Weeks 7–8

- Cost & carbon analysis
 - Dashboard, testing & final evaluation
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10. Expected Outcomes

At the end of the project, the system should:

- Simulate realistic ML workload energy behavior
 - Analyze impact on electrical grids
 - Support distributed workload execution
 - Optimize scheduling and load balancing
 - Evaluate cost, sustainability, and grid stability
 - Provide a scalable and extensible simulation framework
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11. Long-Term Vision

The long-term vision of this project is to develop an intelligent infrastructure-aware scheduling system capable of:

- Energy-aware workload execution
- Carbon-aware optimization
- Distributed HPC coordination
- Real-time decision-making
- Sustainable AI infrastructure management

The project aims to bridge the gap between AI computation, distributed systems, and energy infrastructure analysis.